

Ziliang Xiong, Ph.D. Student

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Education

- 2022 – Now  **Ph.D. specialized in computer vision, Linköping University¹, Sweden**
- Research focus: applying uncertainty quantification into computer vision tasks to improve safety and trustworthiness of autonomous systems, including regression, human pose, end-to-end autonomous driving, etc;
 - Funded by ELLIIT² (Project number: Co8), *Situation Aware Perception for Safe Autonomous Robotics Systems*
 - WASP³ affiliation 2023, including PhD courses, summer school, and study visit
 - Research visit to Delft University of Technology (TU Delft), the Netherlands (Sep 2026 – Feb 2027), funded by the *WASP Research Stints Abroad* program
- 2020 – 2022  **M.Sc. Machine Learning, system and control, Lund University⁴, Sweden**
- GPA: 4.4/5.0
 - Thesis title: *Radar detection using deep learning*.
- 2015 – 2019  **B.Sc. Automation, Beihang University, China**
- Shenyuan honor's degree

Research Publications

- 1 **Z. Xiong**, S. Liu, N. Helgesen, J. Johnander, and P.-E. Forssén, “Catplan: Loss-based collision prediction in end-to-end autonomous driving,” *under review*,  URL: <https://arxiv.org/abs/2503.07425>.
- 2 T. Dehdarirad, M. Felsberg, G. Eilertsen, and **Z. Xiong**, “Commutator-induced uncertainty in VAEs,” *Under review*, 2026.
- 3 S. Liu, **Z. Xiong**, K.-H. Ngo, and P.-E. Forssén, “Matter: Multiscale attention for registration error regression,” *2026 IEEE International Conference on Acoustics, Speech, and Signal Processing*, 2026.
- 4 S. Liu, **Z. Xiong**, B. Wandt, and P.-E. Forssén, “Continuous flow modeling for uncertainty-aware human pose estimation,” *Scandinavian Conference on Image Analysis*, 2025.
- 5 **Z. Xiong**, J. Johanander, and P.-E. Forssén, “Copula calibration for 2d probabilistic human pose estimation,” *Manuscript in preparation*, 2025.
- 6 **Z. Xiong**, A. Jonnarth, A. Eldesokey, J. Johnander, B. Wandt, and P.-E. Forssén, “Hinge-wasserstein: Estimating multimodal aleatoric uncertainty in regression tasks,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, Jun. 2024, pp. 3471–3480.
- 7 **Z. Xiong**, S. Kristoffersson Lind, P.-E. Forssén, and V. Krüger, “Uncertainty quantification metrics for deep regression,” *Pattern Recognition Letters*, vol. 186, pp. 91–97, 2024, ISSN: 0167-8655.  DOI: <https://doi.org/10.1016/j.patrec.2024.09.011>.

¹THE Young University Rankings 31

²ELLIIT is a strategic research environment in information technology and mobile communications, funded by the Swedish government in 2010. With four partners, Linköping University, Lund University, etc, ELLIIT constitutes a platform for both fundamental and applied research, and for cross-fertilization between disciplines and between academic researchers and industry experts.

³The Wallenberg AI, Autonomous Systems and Software Program (WASP) is a major national initiative for strategically motivated basic research, education and faculty recruitment. It is by far the largest individual research program in Sweden.

⁴QS World University Ranking 72

Grants & Projects

- 2022  ***Situation Aware Perception for Safe Autonomous Robotics Systems*** (ELLIIT Project Co8, Linköping University)
- Applying uncertainty quantification to computer vision tasks to improve the safety and trustworthiness of autonomous systems
 - Research areas include regression, human pose estimation, and end-to-end autonomous driving
- 2026  ***Risk Assessment in Vision-Language-Action Models for Automated Vehicles*** (Vinnova ⁵ FFI, Linköping University, 240K SEK)
- Developing risk assessment methods for vision-language-action models in automated driving
 - Funded by Vinnova (Diarienummer: 2026-00800) under the traffic-safe automation program
- 2026.Sep – 2027.Feb  ***WASP Research Stint Abroad*** (TU Delft, the Netherlands, 180K SEK)
- Visiting researcher at Delft University of Technology
 - Funded by WASP to support short-term international research collaboration abroad
- 2021  ***A Vision-based Weed Detection System for the Farmbot*** (Lund University, Robotics Lab)
- Developed a weed detection system that can aid FarmBot to pick up weeds
- Trained YOLO v3 with Darknet on a customized dataset, reaching 83.2% accuracy on the test set
 - Processed and labeled 250+ plant images in 10 classes to build a customized dataset
 - Developed software that drives Farmbot to scan the plants, detect and grip the targets
 - Calibrated the camera and ed the module to project detections to real-world position
-  ***Perspective auto-calibration for security camera applications***(Axis Communication A.B.)
- Developed regression modules to calculate perspective, reducing false positive for object detectors in a post-hoc fashion
 - Developed a novel visualization module to evaluate ML solutions in qualitative and quantitative ways
 - Designed, recorded, and processed videos to form an internal dataset for training and testing regression modules

Employment History

2022. Sep – Now  **Master's Thesis Supervisor, Teaching Assistant, Laboratory Assistant, Linköping University**
- Supervising master's theses, project courses and hosting exercise
2021. Sep – 2021. Nov  **SI-PASS Leader** European Centre for SI-PASS, Lund University
- Hosting supplementary instruction sessions for the course, Image Analysis

⁵Vinnova is Sweden's national innovation agency, funded by the Swedish government to support research and innovation. FFI (Strategic Vehicle Research and Innovation) is a joint programme between the government and the automotive industry focusing on safe and sustainable transport.

Employment History (continued)

2021. Jun – 2021. Sep  **Machine Learning Developer** R&D department, Axis Communication A.B.
Responsible for the project, perspective auto-calibration for security camera applications

Teaching Experience

2023 – Now  Machine Learning for Computer Vision (Master's level); Embedded Perception Systems (Master's level); Signal and System (Bachelor's level), Images and Graphics, Project Course

2021 – 2022  Image Analysis (Master's level)

Mentorship as Primary Advisor

- 2026  Han Xia, **Calibrating Multi-Modal Trajectory Predictions with Pre-rank Regularization: A Case Study on HiVT and Argoverse**, Linköping University
- 2025  Victor Eriksson, **Learned Visibility Filtering for Accurate 2D-to-3D Semantic Projection in LiDAR Point Clouds**, Linköping University
-  Axel Söderlind and Philip Welin-Berger, **YOLO11-LLD: End-to-end object detection and monocular distance estimation for low-light traffic situations**, Linköping University
-  Fredrik Hansson and Lina Tunell, **Video Event Identification using Weakly-Supervised Anomaly Detection**, Linköping University
- 2024  Adam Borgstrand, **Confidence Calibrated Point Cloud Segmentation with Limited Data**, Linköping University
-  Oskar Savinainen, **Uncertainty Estimation and Confidence Calibration in YOLO5Face**, Linköping University
-  Shipeng Liu, **Uncertainty-Aware Human Pose Estimation Using CNFs**, Linköping University
- 2023  Johannes Hägerlind, **3D-reconstruction of the Common Murre**, Linköping University
-  Erik Lidman, **Investigation of Practical Aspects of Multi-Camera Perception for Autonomous Vehicles**, Linköping University

Academic Activity

Peer Review  Reviewer for **International Journal of Computer Vision (IJCV)**, **Pattern Recognition Letters** (3 times), **CVPR 2025**, CVPR workshop **SAIAD** (2024-2026), and **AAAI 2026**

Miscellaneous Experience

Certification

2025, July  **International Computer Vision Summer School**, Spatial Intelligence

2024  **Long Oral Certificate**, Safe Artificial Intelligence for All Domains, CVPR workshop

2023  **Participant Certificate**, WASP Summer School Synthesis of Human Communication

2022  **Affiliated WASP PhD Student positions**, Awarded by Wallenberg AI, Autonomous Systems and Software Program

 **Master's thesis scholarship**, Awarded by Zenseact A.B.

2016 – 2019  **Scholarship for Excellent Study**, Awarded by Beihang University for three years

2017  **Second Prize-China Undergraduate Mathematical Contest in Modeling**. Awarded by China Society for Industrial and Applied Mathematics.

Skills

Languages	📖	Strong reading and writing competencies for English, Mandarin Chinese, Swedish (A1)
Coding	📖	MATLAB, Python, Tensorflow, Pytorch, Git, Julia, SQL, L ^A T _E X, Jax, Apptainer
Misc.	📖	Academic research, teaching, consultation

References

Prof Michael Felsberg

Professor

Linköping University,

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Prof Per-Erik Forssén

Professor

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